Cooling Environments Lightning Talks — This Session

- 9040 — Chemical-Free Water Chemistry Management for Reliable Operation of Copper Cold Plates in Data Center Liquid Cooling
  - Fei Dai — Guangzhou Goaland Energy Conservation Tech. Co., Ltd.
- 9255 — Beyond Chemical Treatment: Advancing Water Quality Management for Reliable Liquid Cooling Infrastructure
  - Stirling Boston — Watts | EasyWater; Kelley Mullick — Avayla Consulting Group

COOLING ENVIRONMENTS

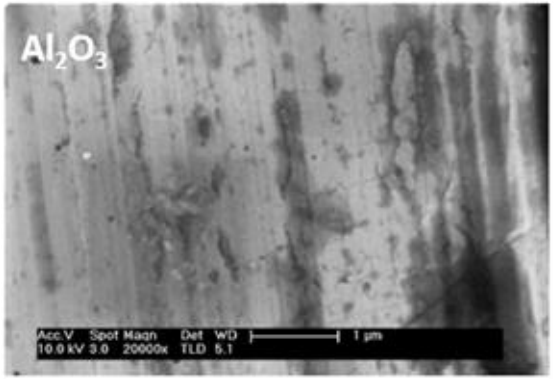
Chemical-Free Water Chemistry Management for Reliable Operation of Copper Cold Plates in Data Center Liquid Cooling

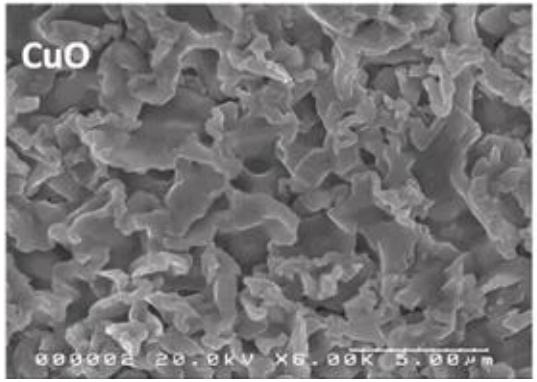
Fei Dai

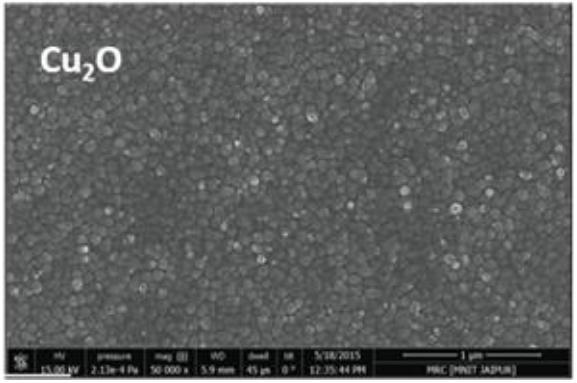
Guangzhou Goaland Energy Conservation Tech. Co., Ltd.

Factors Affecting Corrosion

SEM micrograph of nature film









- Corrosive ions, Cl^- and SO_4^{2-}
- Dissolved oxygen (DO)
- Microbial metabolism

Mat. Lett., **2005**, 59(24-25): 3160-3163.
Mater. Res. Bull., **2023**, 159: 112094.
AIP Publishing LLC, **2016**, 1724(1): 020116.

Specific Impact

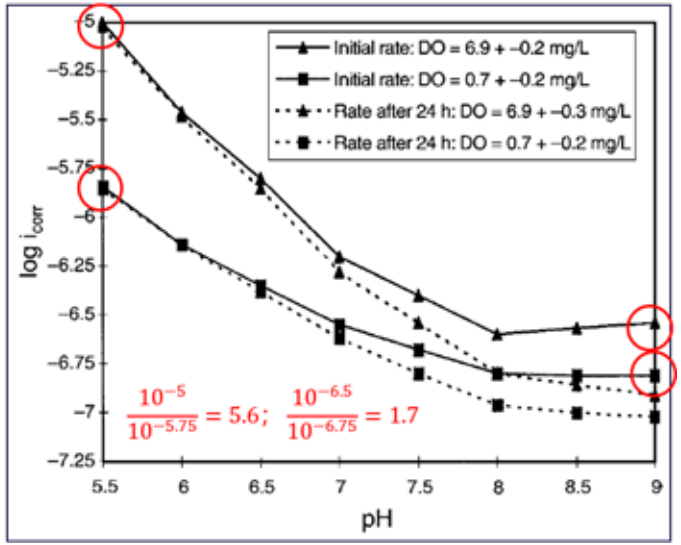
Bengough in the 1920s,

$$Cu^{+} + 2Cl^{-} \Leftrightarrow CuCl_2^{-} + Cl^{-} \Leftrightarrow CuCl_3^{2-} + Cl^{-} \Leftrightarrow CuCl_4^{3-}$$

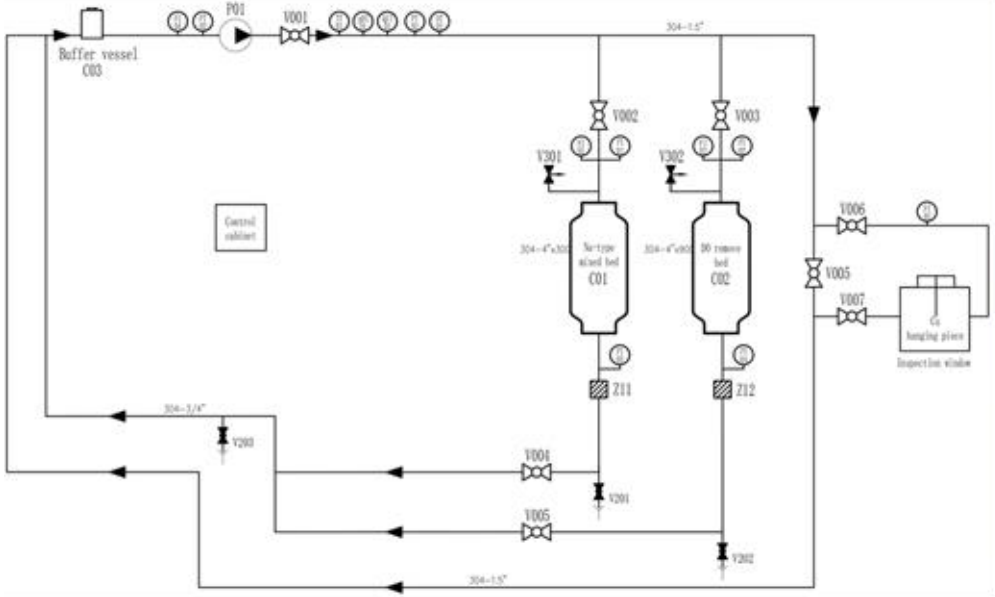
$$K_1 = \frac{[CuCl_2^{-}]}{[Cu^{+}][Cl^{-}]^2} = 10^5 mol^{-2} dm^6$$

Corros. Sci., **2004**, 46.1: 109-135.
J. Inst. Met., **1920**, 23: 65-158.
J. Electrochem. Soc., **1986**, 133: 2035-2043.
Corrosion-NACE., **1978**, 34: 396-406.
Corros. Sci., **1957**, 15: 383-392.
Corros., **2002**, 58.2: 156-165.

- Cl^- : attacks the protective film, coordinates with most metal atoms.
- SO_4^{2-} : plays more of a synergistic role, accelerating corrosion when coexisting with Cl^- or DO.
- CO_2: dissolves in water, can supply H^+, but be controlled by buffering system.
- DO: the corrosion rate increases with rising DO concentration; when DO concentration increases from 0.7 ppm to 6.9 ppm, the corrosion current is amplified by a factor of 1.7 to 5.6.

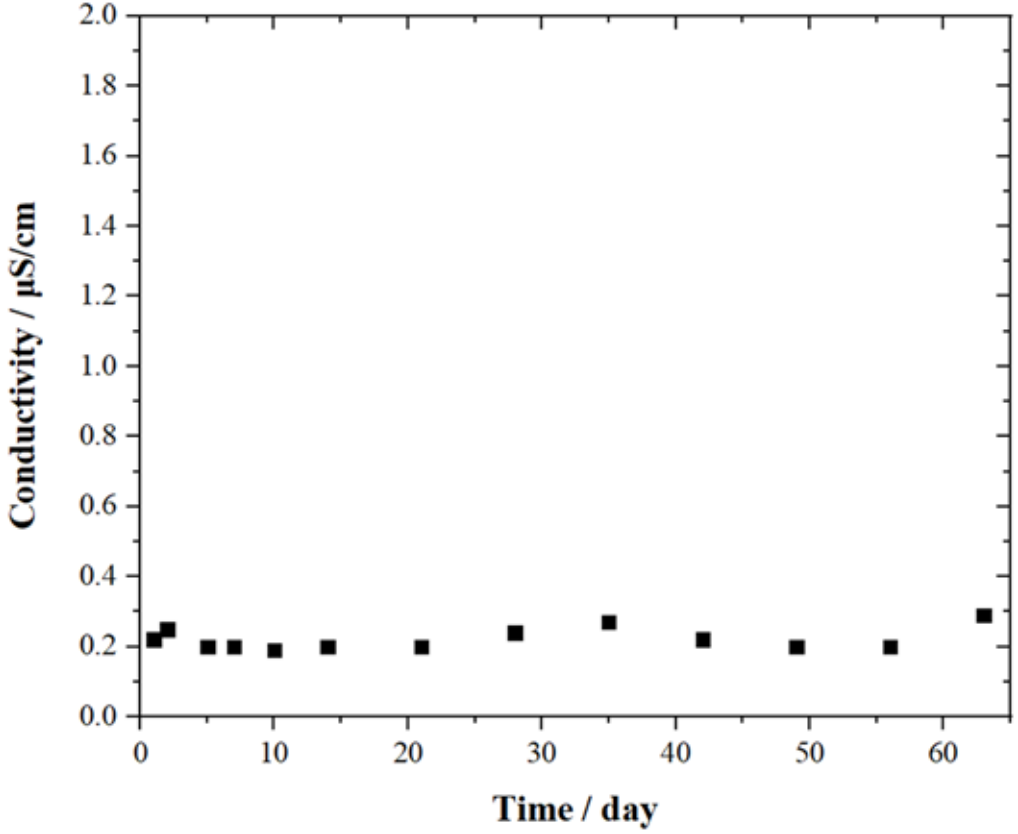


Test Device

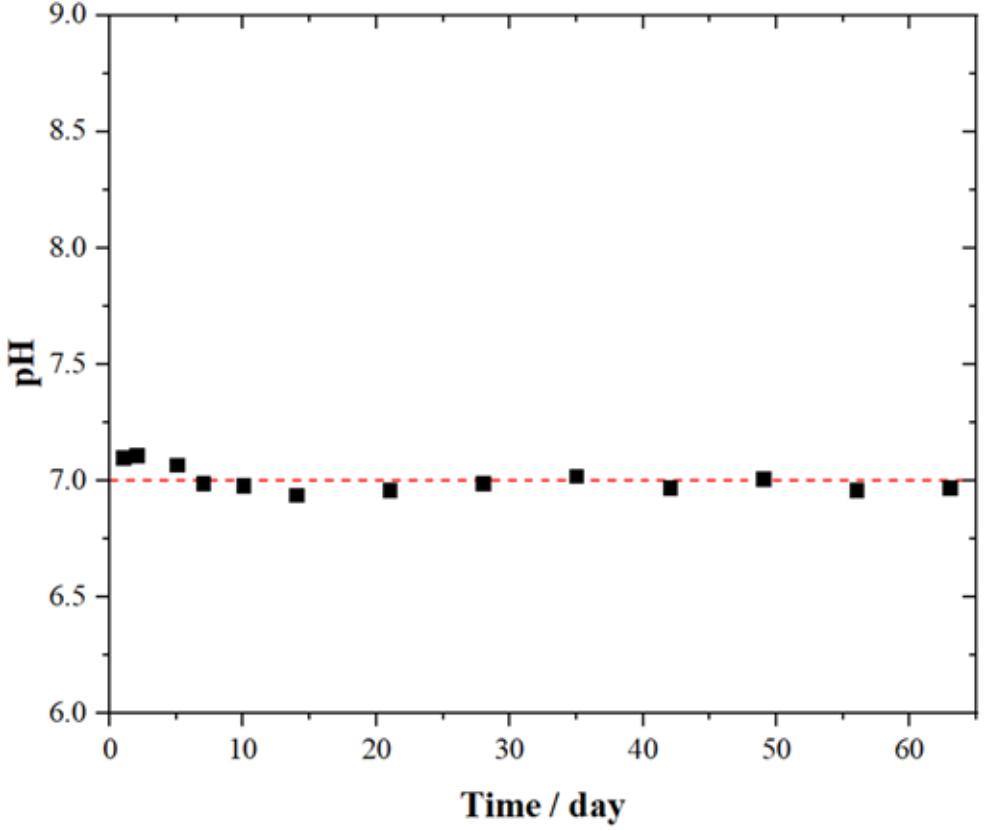

Buffer vessel
C03
P01
V001
304-1.5"
V002
V003
V301
V302
Control cabinet
304-4"x3
Na-type mixed bed
C01
304-4"x30
H2 remove bed
C02
V006
V005
V007
Cu hanging piece
Inspection window
Z11
Z12
304-3/4"
V200
V004
V201
V005
V202
304-1.5"


试验中

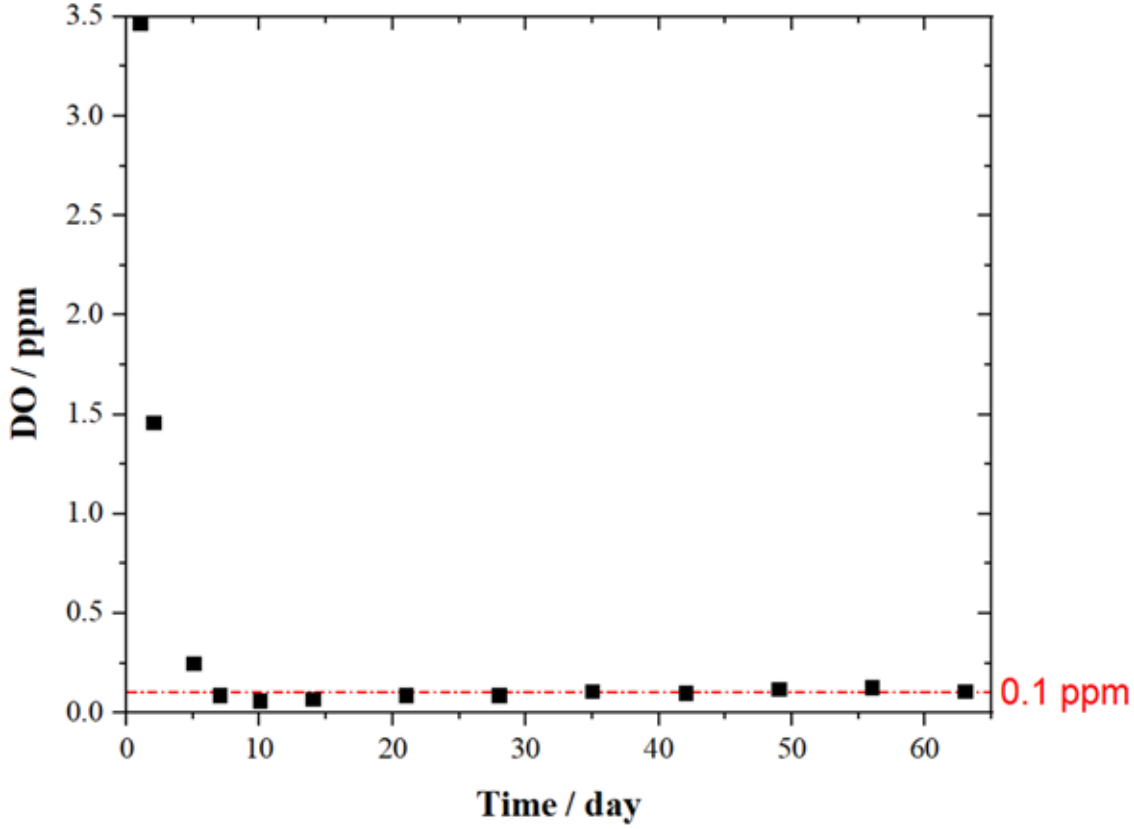
Test Results



Test Results

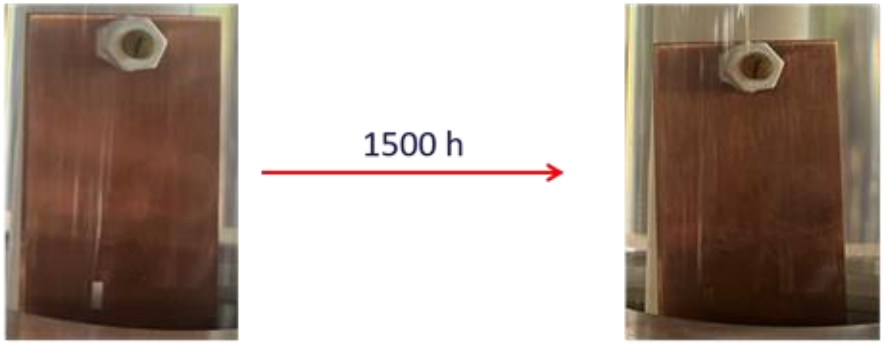


Test Results



Test Results

No.	Weight change / mg	Corrosive rate / μm/y
Sample 1	1.6	0.37
Sample 2	2.3	0.53



- CR (mm/y) = 87.6 × Δm / (S × t × d), where the specimen dimensions are 25.36 × 49.83 × 2.01 mm, giving a surface area S = 28.04 cm^2, test duration t = 1500 h, and material density d = 8.92 g/cm^3.

Conclusion

- Parameters:
  - pH: 6.8-8.5
  - Conductivity ≤ 2.00 μS/cm
  - DO ≤ 0.20 ppm
  - Corrosion rate ≤ 1.00 μm/y
- Next step:
  - Optimization about effluent pH of the resin
  - Parallel tests
  - Pilot project

Product Info

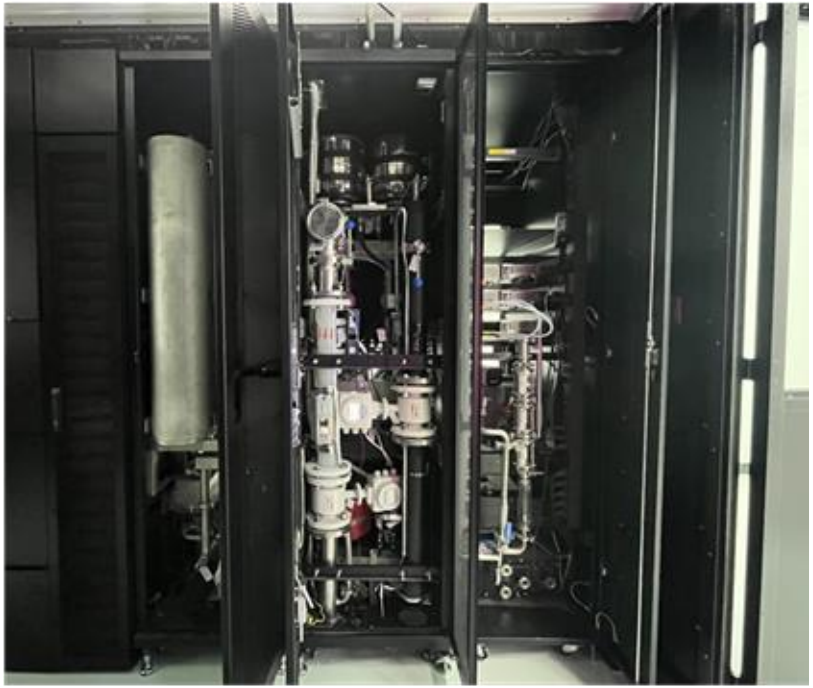
- Photos of product
- Links to Solution Provider's site
  - http://www.goaland.com.cn





Product Info

Item	Parameter
Applicable medium capacity	≤ 6 m^3
Rated flow rate	4–6 m^3/h
Design pressure	4.5 bar
Inlet–outlet differential pressure	≤ 1.2 bar
Rated power	≤ 3 kW
Communication protocol	Modbus RTU / TCP
Control mode	Local / Remote
Human–machine interface	10-inch touch screen
Connection orientation	Top piping, 1.5-inch clamp (ISO 2852)
Noise level	≤ 60 dB (A)
Product dimensions	Depth 1200 mm × Width 600 mm × Height 2200 mm
Layout type	Rack-mountable, accessible from front and rear



Call to Action

- Problem to solve
  - Providing an approach which eliminates the need for chemical additives, reduces waste-fluid generation, simplifies maintenance, and lowers operating costs.
- How to get involved in the project/sub-project community
  - Participate in and Contribute to OCP Cooling Environments workstreams
  - Cold Plate https://www.opencompute.org/wiki/Cooling_Environments/Cold_Plate
  - Advanced Cooling Solution-Cold Plate: https://ocp-all.groups.io/g/OCP-ACS-Cold-Plate


Thank
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San Jose, California

COOLING ENVIRONMENTS

Beyond Chemical Treatment: Advancing Water Quality Management for Reliable Liquid Cooling Infrastructure

Stirling Boston, *Director of Sales, EasyWater | Watts Water Technology*
Kelley Mullick, *CEO, Avayla Consulting Group*

OCP PLATINUM Member

The Reliability Problem: Corrosion Is the Common Failure Mode in Every Closed Loop

- Every closed water loop shares one root vulnerability: corrosion from dissolved oxygen
- Corrosion products drive fouling and become the abrasive debris that erodes downstream components



How Dissolved Oxygen Enters a Nominally “Closed” Loop

- Five common entry points let oxygen into every “closed” loop

1. **Makeup Water**
2. **Vented Tanks**
3. **Faulty Vents**
4. **Shutdowns**
5. **Maintenance**

3–7 ppm

typical dissolved oxygen in an untreated closed loop — enough to sustain continuous corrosion

Why Chemical-Only Treatment Struggles at Cold-Plate Tolerances

- Passivation slows corrosion — it doesn't remove the oxygen driving it
- Sub-micron particulate that chemical-only programs miss is exactly what threatens cold-plate microchannels

10^3 CFU/mL

bacteria threshold — dosing enough biocide to hold this level accelerates corrosion

10^4–10^5 CFU/mL

typical bacteria count in most untreated closed loops

A Water-Chemistry Management Approach: Remove the Driver, Not Just the Symptom

- Continuous side-stream treatment — not periodic batch dosing
- Physical/electronic conditioning — no chemical addition
- Sub-micron (<1 µm) filtration removes iron oxide and sediment
- Removes the driver (O_2 + particulate), not just its symptoms



Field-deployed side-stream: sub-micron filtration + O2/scale

Validation Methodology

- A reproducible four-point protocol for monitoring water chemistry

1. **Dissolved O_2**
   field kit today → online probe next
2. **Corrosion Coupons**
   mils/yr (mpy)
3. **Bacteria**
   CFU/mL dip-slide
4. **Particulate**
   turbidity & TSS

Maps directly onto TCS/CDU water-quality monitoring alongside ASHRAE TC9.9 checks

Closed-Loop Field Results

Dissolved Oxygen — Before / After

Site	Baseline Dissolved O_2	After Treatment
Wolfe County Middle School, KY	5–6 ppm	<1 ppm (2 weeks)
30 Cambridge Park Drive, Boston, MA	Visibly black / turbid	Clear (3 weeks)
Multi-site field average	3–7 ppm (typical baseline)	0–1 ppm (2–3 weeks typical)
Long-term monitored loop	5–6 ppm baseline	<1 ppm sustained (8 months)



Corrosion coupons after treatment: 0.0692 mils/yr measured loss — "excellent" on standard corrosion-rate scale

Why This Matters for Data Center Liquid Cooling

- TCS/CDU loops are closed, copper/brass-wetted systems — the same physics applies directly
- Corrosion particulate adds directly to the erosive contamination that wears high-velocity components
- A chemical-free, monitored program eases tight water-quality tolerances at rack-and-loop scale

≈14%

efficiency loss from a 0.024 in. fouling deposit measured in comparable heat-exchanger tubing — the same physics reaches TCS heat exchangers and cold plates

Key Takeaways

- Dissolved oxygen — not scale alone — drives corrosion, particulate, and fouling in every closed loop
- Chemical dosing alone doesn't stop oxygen ingress or clear sub-micron particulate
- Chemical-free side-stream treatment is field-validated: 3–7 ppm → <1 ppm dissolved O_2 in 2–3 weeks
- A simple four-point protocol gives operators a reproducible way to monitor water quality

What's Next: Field Validation Is Underway

- Third-party lab and client-site testing on a live TCS loop is in progress



Results will be shared back with the OCP community as a full case study.

Call to Action

Problem to solve

- A shared, vendor-neutral water-chemistry validation protocol (dissolved O_2, corrosion rate, bacteria, particulate) for closed-loop TCS/CDU systems with copper cold plates
- Reach out if interested in pursuing a pilot with the system

Join the Cooling Environments workstream discussion on liquid-cooling water-quality specs

For additional information, please contact:

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